

Using dynamic programming for making structured decisions about invasive species

Olivier Gimenez^{1*} Abby Keller^{2*} Lucile Marescot^{3*} Cassie Speakman^{4*}

2026-05-03

¹ CEFÉ, Univ Montpellier, CNRS, EPHE, IRD, Montpellier, France ² address ³ address ⁴ address

* Corresponding author: olivier.gimenez@cefe.cnrs.fr

Invasive alien species (IAS) pose significant ecological, economic, and public health challenges. Robust methods are needed for guiding effective management. Overall, we hope to demonstrate the flexibility and utility of structured decision making in invasive species management.

Target journal: Journal of Animal Ecology

Keywords: Adaptive management, Decision theory, Invasive alien species, Stochastic dynamic programming, Structured decision making, Uncertainty

Introduction

Invasive species are a significant global issue, with wide-ranging impacts on ecosystems, economies, and public health (Pyšek et al. 2020, Roy et al. 2024).

Effective management of invasive species requires blabla for guiding control efforts and evaluating the success of eradication or regulation programs (Williams et al. 2002, Thompson et al. 2021).

Pitch: Structured decision-making (SDM) and adaptive management (AM) have been repeatedly recommended for the management of invasive species, yet they remain underused in practice. There are many reasons for this gap, one of which is the technical challenge of formally identifying and quantifying trade-offs among management objectives. Here, we contribute to ongoing efforts by illustrating the use of stochastic dynamic programming (SDP) and some of its variants — including active and passive adaptive management and partially observable Markov decision processes (POMDPs) — across different ecological contexts (a single population, a metapopulation, and two interacting populations).

Several papers have recommended or implemented structured decision-making (SDM) for managing invasive species. It's worth providing a short review of this literature.

In a recent review on impacts, the use of ML to guide management actions identified as a future research direction (see Appendix 3 in Haubrock et al. (2025)).

“Another recent research direction is the use of machine learning to aid the development of conservation management strategies. These methods involve models that integrate geospatial, climate, species distribution and other data to simulate the distribution of species and vegetation types under different land use and conservation planning scenarios. Reinforcement Learning algorithms can then be used to learn a management policy that attains a specified quantifiable objective (such as minimising the number of individuals from a non-native species) under

constraints such as cost or the amount of land that can be protected. Dietterich, Taleghan and Crowley (2013) demonstrated how Reinforcement Learning can be applied to managing a river system to control the expansion of a non-native plant species. Lapeyrolerie et al. (2022) demonstrated the potential of Reinforcement Learning for resource management problems, such as setting fishing quotas. Conservation area prioritisation through artificial intelligence (CAPTAIN) (Silvestro et al., 2022) is a general simulation framework that allows the user to define flexibly management objectives to be optimised. The unprecedented rate at which new technologies are emerging holds great promise for improving impact assessments related to biological invasions.”

Coypu (*Myocastor coypu*) as a case study throughout the paper. Code and tutorials available as appendices.

Add a shiny for each case study so that folks can play around?

Explanation of the method

“A very brief historical background. Why is the method useful for investigating or analysing the topic in question? Provide some details on the method’s applications.”

We do not want to rewrite Lucile’s paper...

Invasive species are a major driver of ecological disruption, economic losses, and risks to human health and food security, and their management often requires repeated decisions under uncertainty, limited budgets, and imperfect knowledge of ecological dynamics, which are frequently characterized by strong population fluctuations, rapid spread, and broad spatial distributions. In practice, managers must decide not only whether to intervene, but also when, where, and how strongly, while balancing short-term operational costs against long-term socio-ecological benefits (Hauser and Possingham 2008). This makes invasive species management a

complex decision problem, in which predicting invasion dynamics, or risk forecasting alone are insufficient unless it is explicitly linked to management objectives, feasible actions, and measurable consequences.

Although invasive species management does not always involve strongly opposed objectives, as in some human–wildlife conflicts, the utility function can still be highly complex. Management decisions must weigh expected effectiveness against economic costs, operational effort, feasibility, timing, and the potential unintended impacts of control actions, including non-target effects (Poulin 2015). These decisions are further complicated by the fact that invasive species and their control may be perceived differently by different stakeholders, so that social acceptability becomes an important component of management outcomes (Estevez et al. 2014). More broadly, controversies surrounding invasive species management may reflect differences in environmental values, risk perceptions, rather than disagreement over ecological evidence alone (Simberloff 2012). In this context, several authors have argued that assessing ecological effectiveness of invasive species control measures is not sufficient, and that management strategies should also be evaluated in terms of their social impacts (Crowley et al. 2017). Recent studies further confirm that commonly proposed management interventions, particularly those involving lethal control, often face substantial public resistance or stakeholder conflict, making social feasibility a central dimension of management success (Simmons et al. 2025; Busby et al. 2025). That’s why SDP is an appropriate method for invasive species. . . In invasive species management, SDP has been applied to problems such as balancing prevention and control for zebra mussels, optimizing search effort for red imported fire ants in Australia, selecting integrated control strategies for invasive weeds such as leafy spurge or blackberry, and determining whether prevention, eradication, control, or adaptation is optimal under different stages of aquatic invasions.

In practice, SDP treats invasive species management as a sequential decision problem in which

actions are optimized over time rather than evaluated in isolation. The sequence of decisions are optimal actions over time that depend on the current state of the system, such as infestation size, invaded area, local density. At each time step, managers can choose among several actions including preventing establishment, conducting routine surveillance, intensifying search in high-risk areas, containing spread from invaded sites, reducing local abundance through control, attempting eradication of already established populations, or postponing intervention when action is not expected to be worthwhile (Moore et al. 2011). The consequences of each action are then evaluated under stochastic population dynamics, imperfect detection (in the case of POMDP, e.g Haight & Polasky (2010)) and partial controllability, while accounting for the expected future effects of each action on progressing towards the ecological, economic, or social objectives that management seeks to optimize (Williams et al. 2001). More specifically, SDP has been used to determine how much to invest in prevention versus post-invasion control for the invasive zebra mussels within nuclear power-plant water systems (Leung et al. 2002), to optimize control strategies (mechanical, chemical, biological) for the invasive weed leafy spurge by combining information on density and spatial spread (Hyder et al. 2008), to identify state-dependent management policies for an invasive gypsy moth (*Lymantria dispar*) considering the metapopulation with different patch sizes and spread dynamics (Bogich and Shea 2008), to optimize search effort for the red imported fire ant in Queensland by balancing broad surveillance against targeted search and learning (Baxter et al. 2011), and to compare prevention, control, eradication, and adaptation strategies for aquatic invasive species under different stages of invasion (Hyytiäinen et al. 2013).

Structured decision making

Why and what (all)

This is the expected discounted future value; if I take action a now in some state, this is my average (hence the sum) of tomorrow's best-possible rewards, scaled down by β because tomorrow's payoffs are worth less than

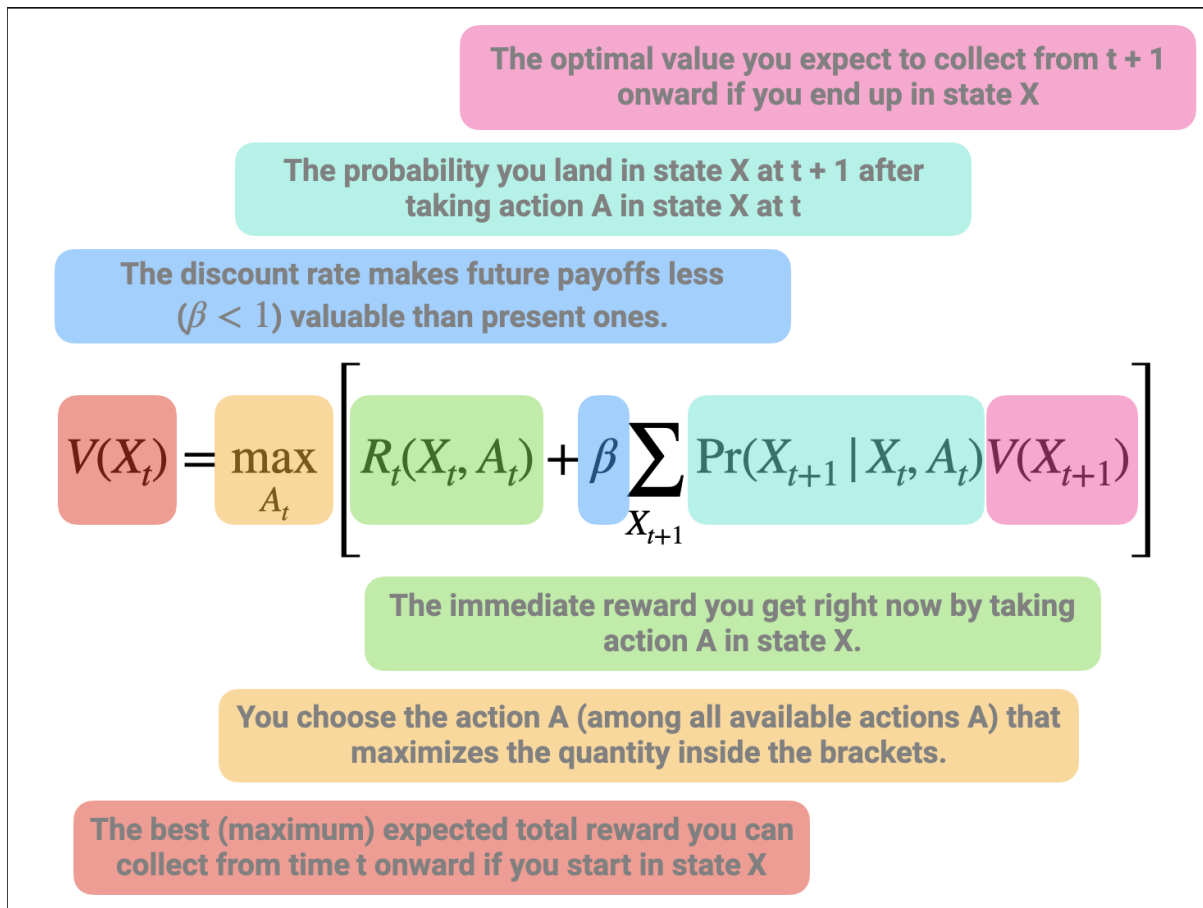
⊕

$$V(X_t) = \max_{A_t} \left[R_t(X_t, A_t) + \beta \sum_{X_{t+1}} \Pr(X_{t+1} | X_t, A_t) V(X_{t+1}) \right]$$

The immediate reward you get right now by taking action A in state X .

You choose the action A (among all available actions A) that maximizes the quantity inside the brackets.

The best (maximum) expected total reward you can collect from time t onward if you start in state X



110

111 Here we present a minimal stochastic dynamic programming (SDP) example. The aim is purely
 112 pedagogical: to show how states, actions, transition probabilities, rewards, and time horizon
 113 combine to produce an optimal decision rule.

114 We consider a stylized invasion management problem with three qualitative states describing
 115 the invasion level:

- 116 • Eradicated: no individuals detected (possible re-invasion risk)
- 117 • Contained: invasion present but at low abundance
- 118 • Established: invasion widespread with high impacts

119 At each time step, the manager chooses between two actions:

- 120 • Monitor: surveillance only (low cost, little effect)
- 121 • Control: active management campaign (higher cost, improves state)

122 State dynamics are described by action-dependent transition matrices

$$P_a(s, s') = \Pr(S_{t+1} = s' \mid S_t = s, A_t = a).$$

123 Transition matrices describe how the invasion state evolves probabilistically from one time
124 step to the next under each management action. Each row corresponds to the current state and
125 each column to the next state, with entries giving transition probabilities. We build a transition
126 matrix for each action.

127 Under monitoring, the transition is:

		Erad.	Cont.	Estab.
$P_{\text{Monitor}} =$	Erad.	0.95	0.05	0.00
	Cont.	0.00	0.80	0.20
	Estab.	0.00	0.00	1.00

128 If the system is eradicated, it remains so with probability 0.95 but may revert to a contained
129 state with probability 0.05, reflecting a small risk of re-invasion. When the invasion is contained,
130 it persists with probability 0.80 but may deteriorate to an established state with probability
131 0.20, indicating possible population growth without intervention. Once established, the system
132 remains established with probability 1, meaning monitoring alone cannot reverse the invasion.

133 In contrast, control shifts the system toward better states:

		Erad.	Cont.	Estab.
$P_{\text{Control}} =$	Erad.	0.98	0.02	0.00
	Cont.	0.20	0.75	0.05
	Estab.	0.05	0.75	0.20

134 From an eradicated state, the system remains eradicated with probability 0.98 and only rarely
135 becomes contained (0.02). When contained, control can eradicate the population (0.20), maintain
136 it at low levels (0.75), or fail and allow establishment (0.05). Finally, when the invasion is
137 established, control improves the situation in most cases, leading to a contained state with
138 probability 0.75, occasionally achieving eradication (0.05), but sometimes failing to reduce
139 impacts (remaining established with probability 0.20).

140 Together, these transition matrices encode both ecological uncertainty and the imperfect effec-
141 tiveness of management actions.

142 Next we define rewards. Rewards are defined as the negative of total costs, so that maximizing
143 expected reward is equivalent to minimizing management and damage costs.

$$R(s, a) = -(\text{damage}(s) + \text{cost}(a)).$$

144 Damage depends on the invasion state (higher when the invasion is established), while manage-
145 ment cost depends on the chosen action (higher for active control than monitoring). The reward
146 matrix is constructed by combining these two components for all state–action pairs, yielding
147 the immediate payoff used in the dynamic programming recursion.

		Monitor	Control
$R =$	Erad.	−1	−6
	Cont.	−5	−10
	Estab.	−15	−20

148 (Rows correspond to the current invasion state and columns to the management action.)

149 For example, when the invasion is eradicated, monitoring yields a reward of -1, reflecting
150 only minor surveillance costs, whereas applying control in the same state yields -6 due to

higher intervention costs. When the population is contained, rewards are -5 for monitoring and -10 for control, representing moderate damages combined with the respective management costs. Finally, when the invasion is established, rewards drop to -15 under monitoring and -20 under control, reflecting both high ecological impacts and, in the latter case, substantial intervention effort. Overall, this reward structure captures the short-term trade-off between ecological impacts and management effort that underlies the decision problem.

Solving the finite-horizon decision problem yields the optimal value function $V_t(s)$, which represents the expected cumulative reward (i.e., negative total cost) obtained when starting from state s at time t and following the optimal policy thereafter. Values are more negative when the invasion is established, reflecting higher expected damages. Moreover, values increase (become less negative) as the horizon shortens because fewer future costs can accumulate.

	$t = 1$	$t = 2$	$t = 3$	$t = 4$	$t = 5$	$t = 6$
$V_t(s) =$						
$s = \text{Erad.}$	-7.60	-5.50	-3.69	-2.20	-1	0
$s = \text{Cont.}$	-33.78	-27.43	-19.96	-12.0	-5	0
$s = \text{Estab.}$	-49.26	-42.05	-34.47	-26.8	-15	0

The associated optimal policy $\pi_t(s)$ specifies the best action to take in each state s at each time step t . Overall, the policy indicates that monitoring is typically optimal when the system is eradicated, whereas control is recommended when the invasion is established due to the high expected damages. In the contained state, the optimal decision varies with the time horizon, reflecting the trade-off between immediate intervention costs and the risk of future deterioration.

	$t = 1$	$t = 2$	$t = 3$	$t = 4$	$t = 5$
$\pi_t(s) =$					
$s = \text{Erad.}$	Monitor	Monitor	Monitor	Monitor	Monitor
$s = \text{Cont.}$	Control	Control	Monitor	Monitor	Monitor
$s = \text{Estab.}$	Control	Control	Control	Control	Monitor

168 This toy example highlights key features of SDP:

- 169 • **State dependence:** optimal decisions depend on invasion level.
- 170 • **Trade-off:** control is costly but reduces future risk.
- 171 • **Time horizon effects:** as the horizon shortens, actions may become more short-term
- 172 oriented.
- 173 • **Uncertainty:** decisions account for probabilistic transitions, not deterministic outcomes.

174 Things to consider for before using this method

175 “Outline the requirements for framing hypotheses, data and computing, experimental constraints
176 and statistical power.”

177 Worked examples

178 “Papers should include at least two tutorials, with a step-by step demonstration of the method.
179 Figures and mathematical formulations are particularly useful here. Where possible, tutorials
180 and examples should include those using the R statistical package or other freeware.”
181 Here explain that we use coypru IAS in Europe as a “fil rouge”.

Single population (Olivier)

The objective of this example is to determine an optimal regulation policy for coypu based on their total abundance. We consider a stylized management problem in which a decision maker chooses, at each time step, the intensity of a control effort $u \in [0, 1]$. Low effort corresponds to minimal intervention, while high effort represents intensive removal. The management objective is to balance three competing components: ecological damages that increase with population size (e.g. crop losses, dike weakening, nuisance), management costs that increase with effort, and an additional penalty if abundance exceeds a tolerable level.

In contrast to the previous example with discrete invasion states, the system is here described by a continuous state variable, the population size N . Because dynamic programming requires a finite state space, abundance is discretized onto a grid ranging from 0 to the carrying capacity K . In the implementation, we set $K = 2000$ individuals and use a grid step of 20 individuals. Control effort is also discretized, from 0 to 1 in increments of 0.05.

Population dynamics follow a density-dependent growth model with removals due to control,

$$N_{t+1} = N_t + rN_t \left(1 - \frac{N_t}{K}\right) - q(u)N_t,$$

where the first term describes logistic growth and $q(u)$ is the fraction of individuals removed as a function of effort. We use a somehow saturating removal function,

$$q(u) = 1 - \exp(-\alpha u),$$

with $\alpha = 1.5$, which implies diminishing returns of control effort (Figure 1). For example, an effort of $u = 0.5$ removes about 53% of the population, whereas maximum effort $u = 1$ removes about 78%, reflecting the fact that complete eradication is rarely achievable in a single step.

Rewards are defined as the negative of total costs, so that maximizing expected reward is

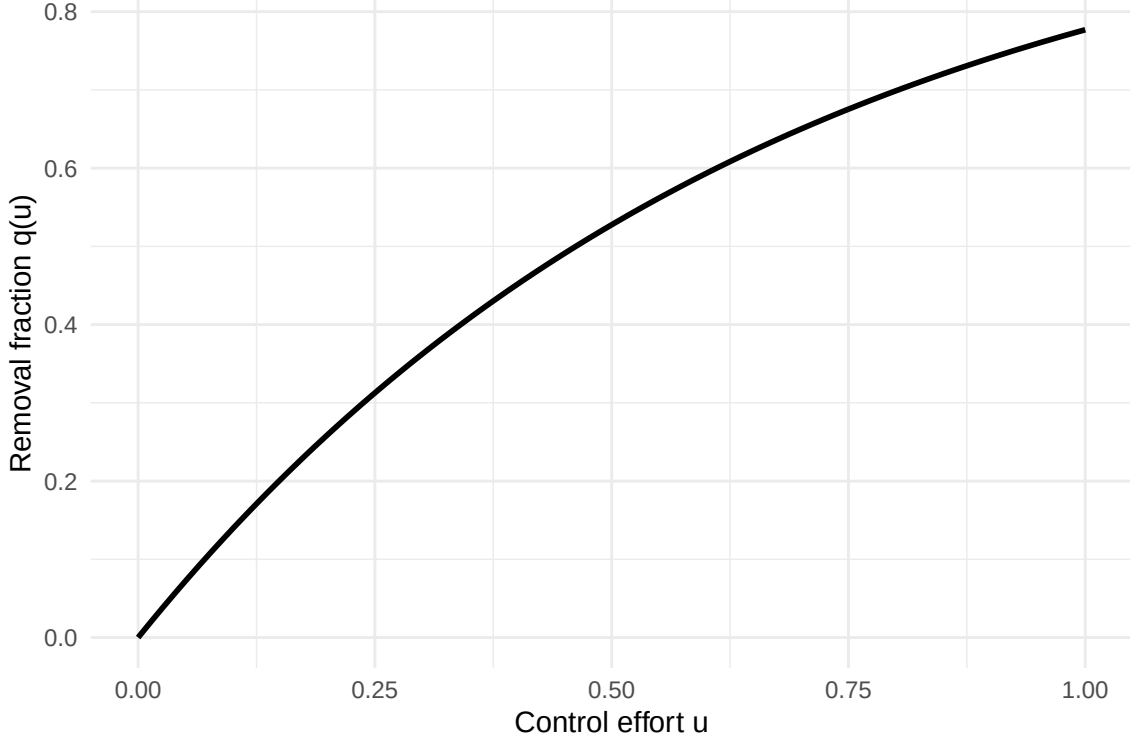


Figure 1: Removal fraction $q(u)$ as a function of control effort u .

equivalent to minimizing long-term damages and management expenses (Figure 2):

$$R(N, u) = -(\text{damage}(N) + \text{cost}(u) + \text{penalty}(N)).$$

Damages increase non-linearly with abundance, here using a simple quadratic form $\text{damage}(N) = 0.1N + 0.001N^2$, reflecting the common ecological pattern whereby impacts accelerate at high density. Management costs increase with effort and are assumed convex ($\text{cost}(u) = 50u + 200u^2$), capturing increasing marginal costs of intensive interventions. Finally, a penalty is applied when abundance exceeds a tolerable threshold $N_{\text{tol}} = 400$, representing situations where ecological or social impacts become unacceptable. The penalty increases linearly beyond this threshold according to

$$\text{penalty}(N) = \begin{cases} 0, & N \leq N_{\text{tol}}, \\ 2000 \frac{N - N_{\text{tol}}}{N_{\text{tol}}}, & N > N_{\text{tol}}. \end{cases}$$

210 This formulation implies that the penalty is zero below the threshold and then rises proportion-
 211 ally to the exceedance. For example, an abundance of $N = 800$ yields a penalty of 2000, thereby
 212 strongly incentivizing control once populations grow beyond acceptable levels.

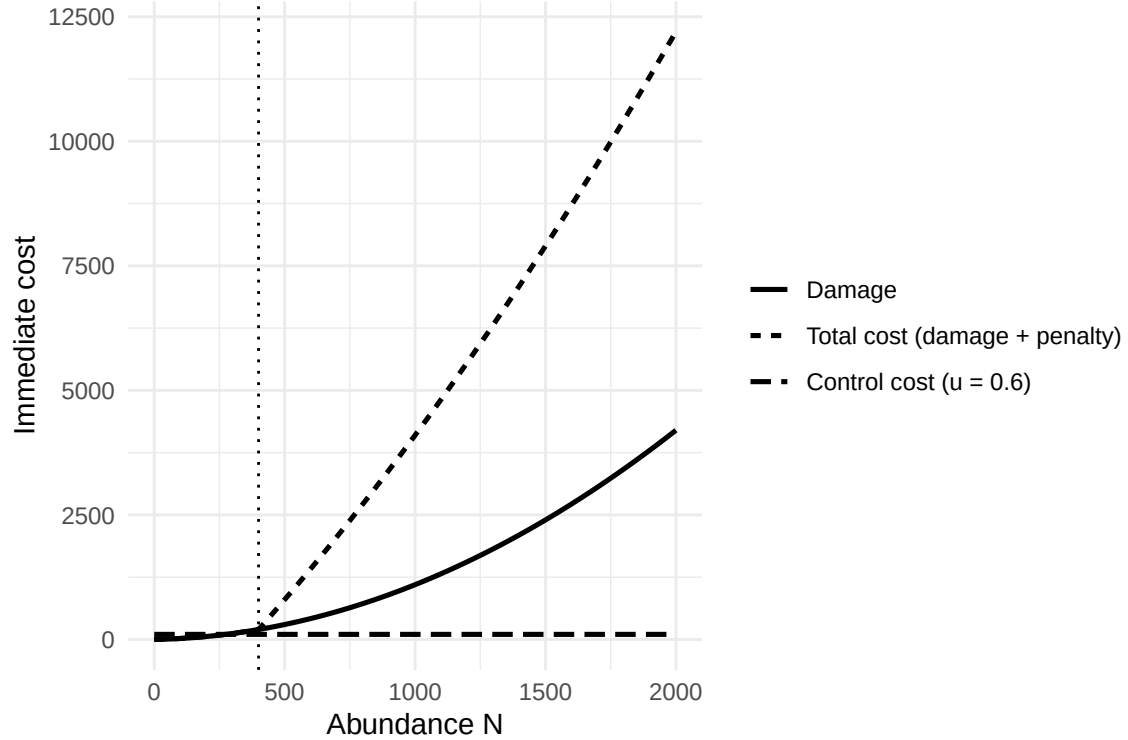


Figure 2: Immediate cost components as functions of abundance. Damage increases nonlinearly with population size, leading to rapidly rising total costs (damage + penalty). The dotted line marks the tolerable threshold ($N_{\text{tol}} = 400$) above which an additional penalty is applied in the decision model, although it is not shown here for clarity. Control cost is illustrated for an example effort level ($u = 0.6$).

213 The decision problem is solved using infinite-horizon discounted value iteration (with discount
 214 factor $\gamma = 0.99$). The resulting value function represents the expected long-term net benefit of
 215 starting from each abundance level when optimal decisions are applied thereafter. The optimal
 216 policy $u(N)$ typically exhibits a threshold-like pattern: low effort at low abundance where
 217 damages are limited, progressively stronger intervention as abundance increases, and intensive
 218 control once the population approaches or exceeds the tolerable level. This behavior reflects a
 219 general ecological intuition — management intensity should scale with both population size
 220 and risk — and illustrates how stochastic dynamic programming formalizes this principle
 221 within a transparent decision framework.

The decision problem is solved using infinite-horizon discounted value iteration (discount factor $\gamma = 0.99$). This yields (i) an optimal stationary policy, mapping each abundance class to a recommended control effort, and (ii) a value function (not shown) that summarizes the long-term expected net benefit when the policy is followed.

Because damages and the penalty increase sharply with abundance, the optimal rule is “threshold-like” (top panel in Figure 3). Effort is low at small N , then increases rapidly as N approaches the tolerable threshold $N_{\text{tol}} = 400$, and remains high for large N . Ecologically, this tells us that when the population is small, impacts are limited and additional control has diminishing returns, whereas once abundance enters an unacceptable zone, strong intervention is warranted to prevent sustained high impacts.

The bottom panel of Figure 3 illustrates the dynamical consequences of applying this rule to a trajectory starting from $N_0 = 800$. Control effort $u(t)$ adjusts immediately to the current abundance class, and abundance $N(t)$ responds through the logistic dynamics with removals. In this deterministic grid-based version, the system typically moves quickly toward a lower-abundance regime and then stabilizes around a band where the marginal benefit of additional control balances its marginal cost.

Metapopulation (Olivier)

We extend the single-population coyote example to a spatially structured invasion in which several sites are connected through dispersal. The aim is to explore how management effort should vary through time when local demographic processes and spatial spread interact. We consider a network of several sites that can be interpreted as cities or management units. At each monthly time step, a decision maker chooses a global management action corresponding to a harvest intensity applied simultaneously across all sites. Local populations then grow according to density dependence, are partially removed by management, and may subsequently

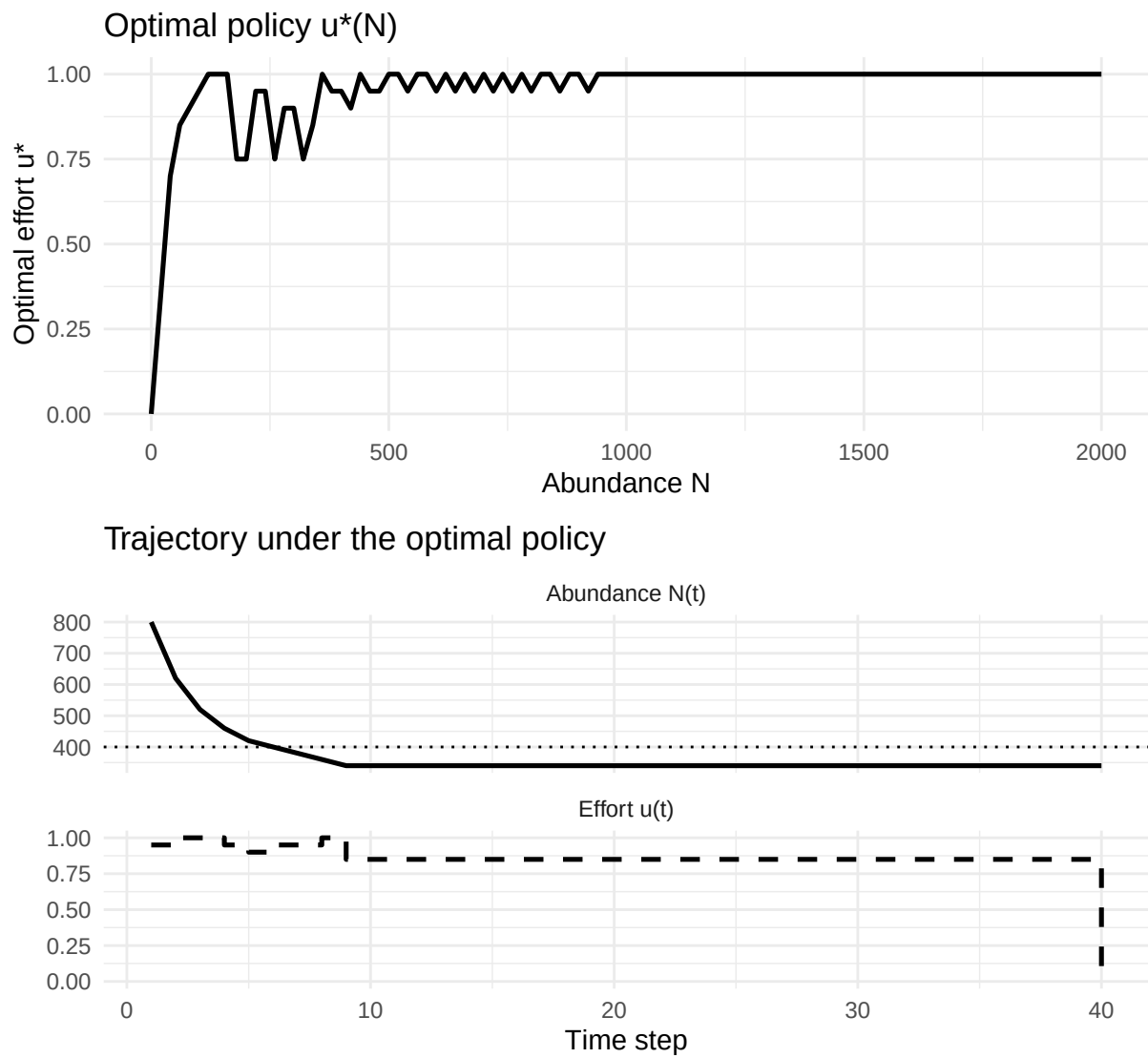


Figure 3: Optimal policy $u^*(N)$ (top) and an example trajectory under the optimal policy (bottom, faceted into abundance and effort), starting from $N_0 = 800$.

undergo stochastic extinction or colonization depending on local abundance and the occupancy of neighboring sites.

At time t , each site i is either occupied or empty, represented by the binary variable $X_i(t) \in \{0, 1\}$. When a site is empty its abundance is zero, whereas when it is occupied its population size follows discrete logistic growth with intrinsic rate r and carrying capacity K . Denoting abundance by $N_i(t)$, post-growth abundance is

$$N_i^{\text{grow}}(t+1) = N_i(t) + r N_i(t) \left(1 - \frac{N_i(t)}{K}\right).$$

Management then removes a fraction h_a of the population, so that

$$N_i(t+1) = N_i^{\text{grow}}(t+1) (1 - h_a).$$

This formulation captures the idea that control reduces numbers but does not eliminate underlying demographic regulation.

Occupancy is updated stochastically based on local abundance and spatial context (Figure ??).

When a site is occupied, extinction probability decreases with abundance according to

$$p_{\text{ext}}(N_i) = \exp(-\alpha N_i),$$

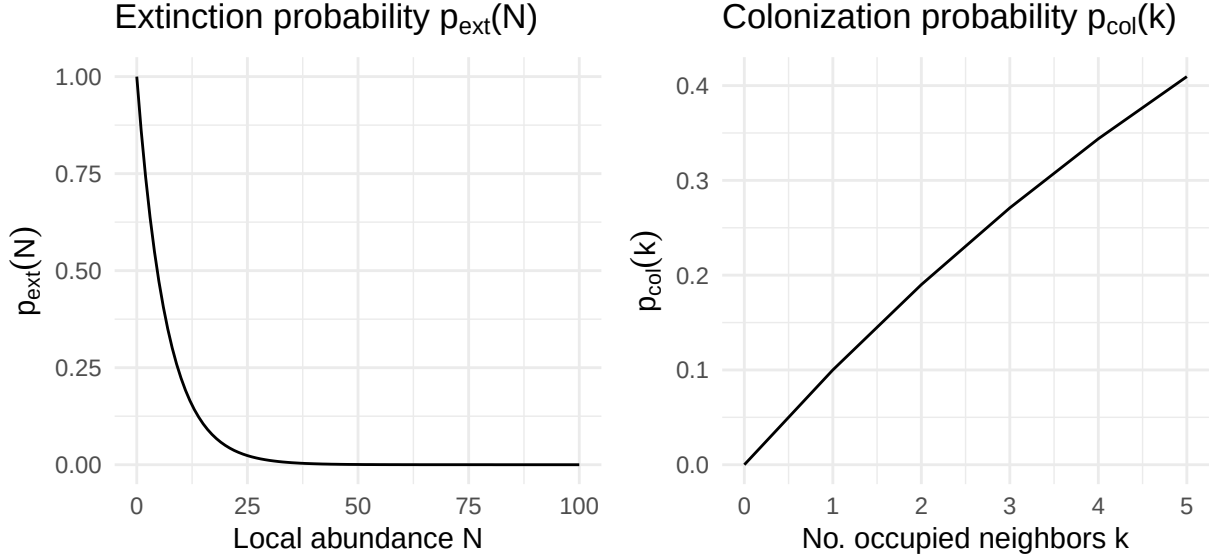
so that small populations are more vulnerable to disappearance. When a site is empty, colonization probability increases with the number of occupied neighbors

$$k_i(t) = \sum_{j \neq i} A_{ij} X_j(t),$$

through

$$p_{\text{col}}(k_i) = 1 - (1 - \beta)^{k_i}.$$

Together, these rules link local demography with spatial dynamics and create feedbacks between sites.



Because the transition probabilities induced by these ecological rules are analytically intractable, we approximate them using repeated simulations. Starting from a given state and management action, we simulate the ecological dynamics many times and record the frequency of each possible next state. In practice, the estimated probability of transitioning from state s to state s' under action a is the proportion of simulations that lead to s' :

This Monte Carlo approach allows the decision model to incorporate nonlinear and stochastic ecological processes without requiring an explicit analytical expression for transition probabilities.

Management decisions are evaluated through a cost function that balances ecological damages and intervention effort (Figure 4). At each time step, total cost is defined as

$$C(s, a) = c_d \sum_{i=1}^n N_i(t) + c_a(a),$$

where c_d represents the damage cost per individual (here $c_d = 1$) and $c_a(a)$ is the management cost associated with action a . In this example, we consider three possible actions corresponding to no control, moderate control, and strong control, with costs

$$c_a(a) \in \{0, 5, 10\}.$$

The reward used in the dynamic programming recursion is defined as the negative of this cost,

$$R(s, a) = -C(s, a),$$

so that maximizing expected reward is equivalent to minimizing long-term total costs.

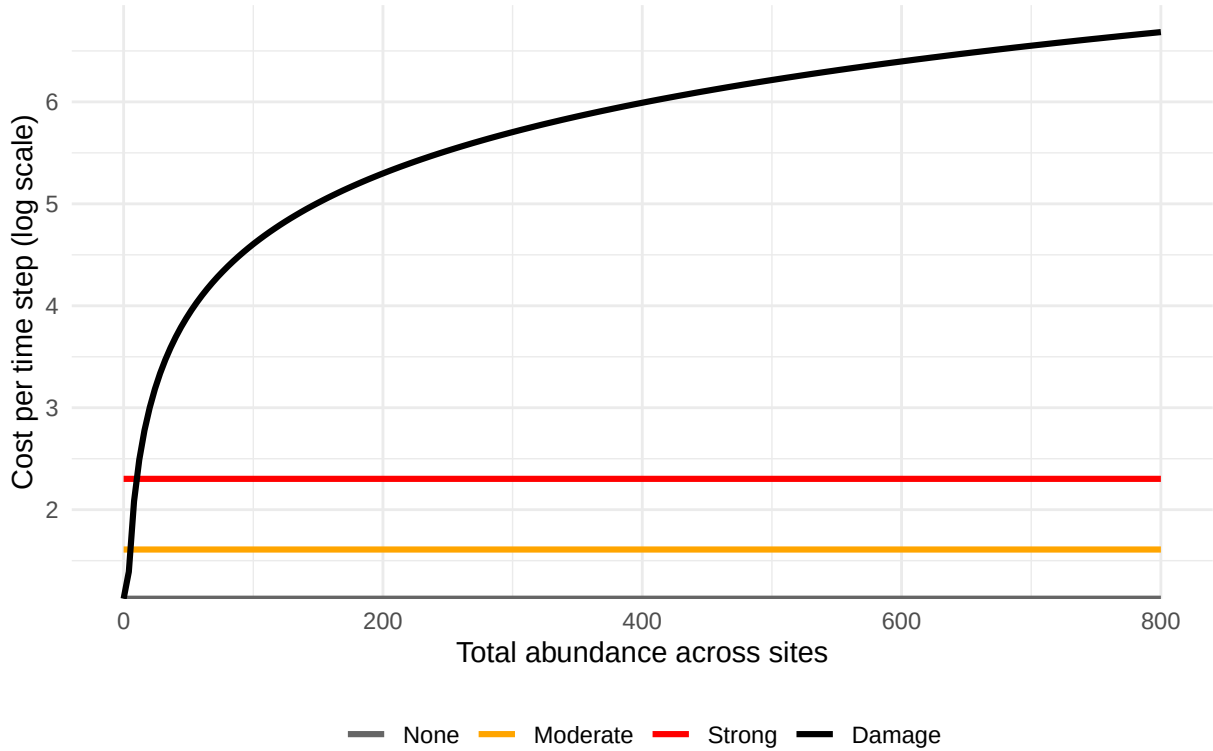


Figure 4: Representation of the cost components on the log scale. Damage costs increase linearly with total abundance across sites, whereas control costs are fixed action-specific intercepts (none/moderate/strong). Stronger control increases immediate spending but can reduce future abundance and damages.

The resulting policy is illustrated in Figure 5. When few sites are occupied and populations remain small, limited intervention is often sufficient because damages are modest and extinction

may occur naturally. As more sites become occupied or local abundances increase, stronger control becomes optimal in order to reduce future spread and prevent escalating impacts. Temporal variation in recommended actions arises because the finite planning horizon reduces the benefit of aggressive intervention late in the time period.

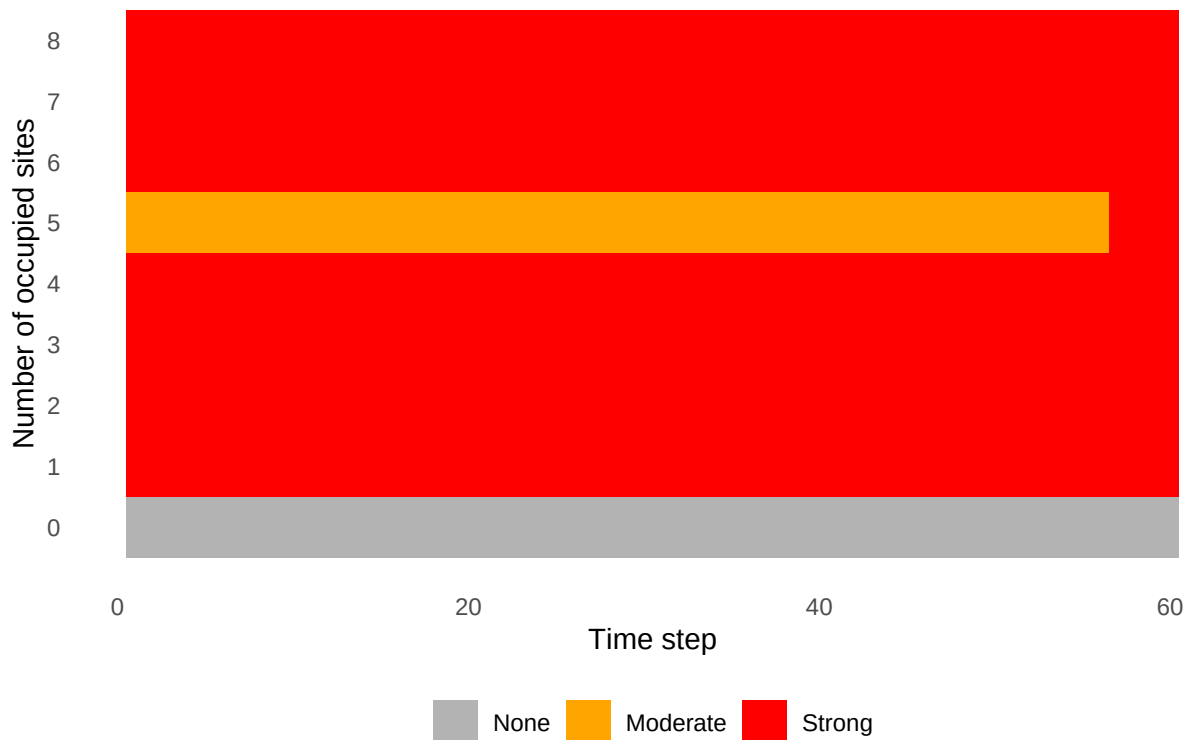


Figure 5: Optimal global action as a function of time and invasion level (measured as the number of occupied sites). Colors indicate the action recommended by the finite-horizon SDP (None/Moderate/Strong). This visualization highlights how stronger control becomes optimal as the invasion occupies more sites, and how recommendations may soften near the end of the planning horizon.

Comparing forward simulations under the optimal policy with simulations under constant strategies highlights the value of adaptive management (Figure 6). The optimal strategy typically reduces cumulative costs by applying effort strategically in time rather than maintaining a fixed level of control.

Spatial trajectories further illustrate how coordinated management alters invasion dynamics across the network, with reductions in abundance and occupancy emerging progressively across sites (Figure 7).

In perspective, say that this can be used to model competition or predator-prey, or host-pathogen

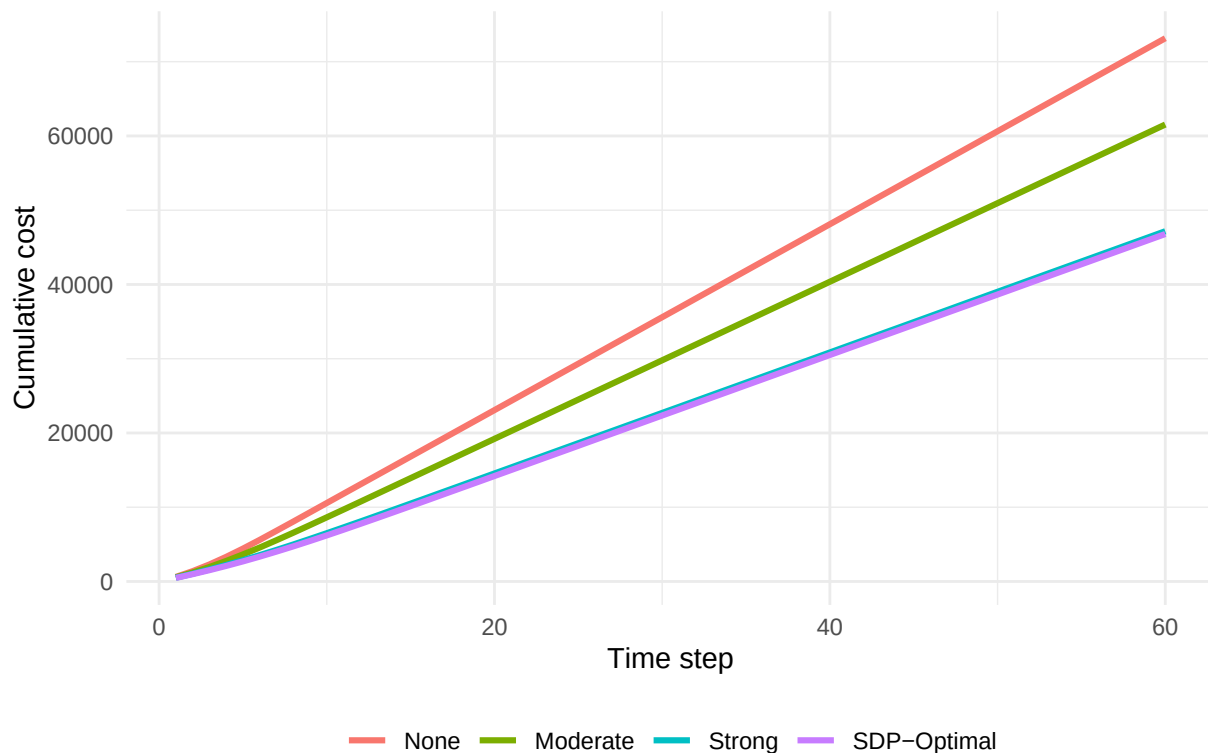


Figure 6: Cumulative costs over time under constant strategies (none, moderate, strong) versus the SDP-optimal policy. Costs combine damages proportional to total abundance and an action-specific control cost.

292 / disease thing. See SDP-species-interactions/ in Olivier biblio repo.

293 **Accounting for observation error w/ POMDP (Cassie)**

294 Illustrate POMDP.

295 <https://besjournals.onlinelibrary.wiley.com/doi/10.1111/2041-210X.14374> <https://besjournals.onlinelibrary.wiley.com/doi/10.1111/2041-210X.13692>

296 [onlinelibrary.wiley.com/doi/10.1111/2041-210X.13692](https://besjournals.onlinelibrary.wiley.com/doi/10.1111/2041-210X.13692)

297 In an ideal world, invasive species management would be based on perfect knowledge of the
 298 state of the system. Yet ecological systems are inherently difficult to monitor, leading to imperfect
 299 knowledge of the system and its state. While ongoing monitoring can reduce uncertainties
 300 associated with the state of the system, invasive species management often requires timely
 301 decisions. In these cases, partially observable Markov decision process (POMDP) models can
 302 be used. POMDPs build on MDPs, dealing with uncertainty in both the model and state of

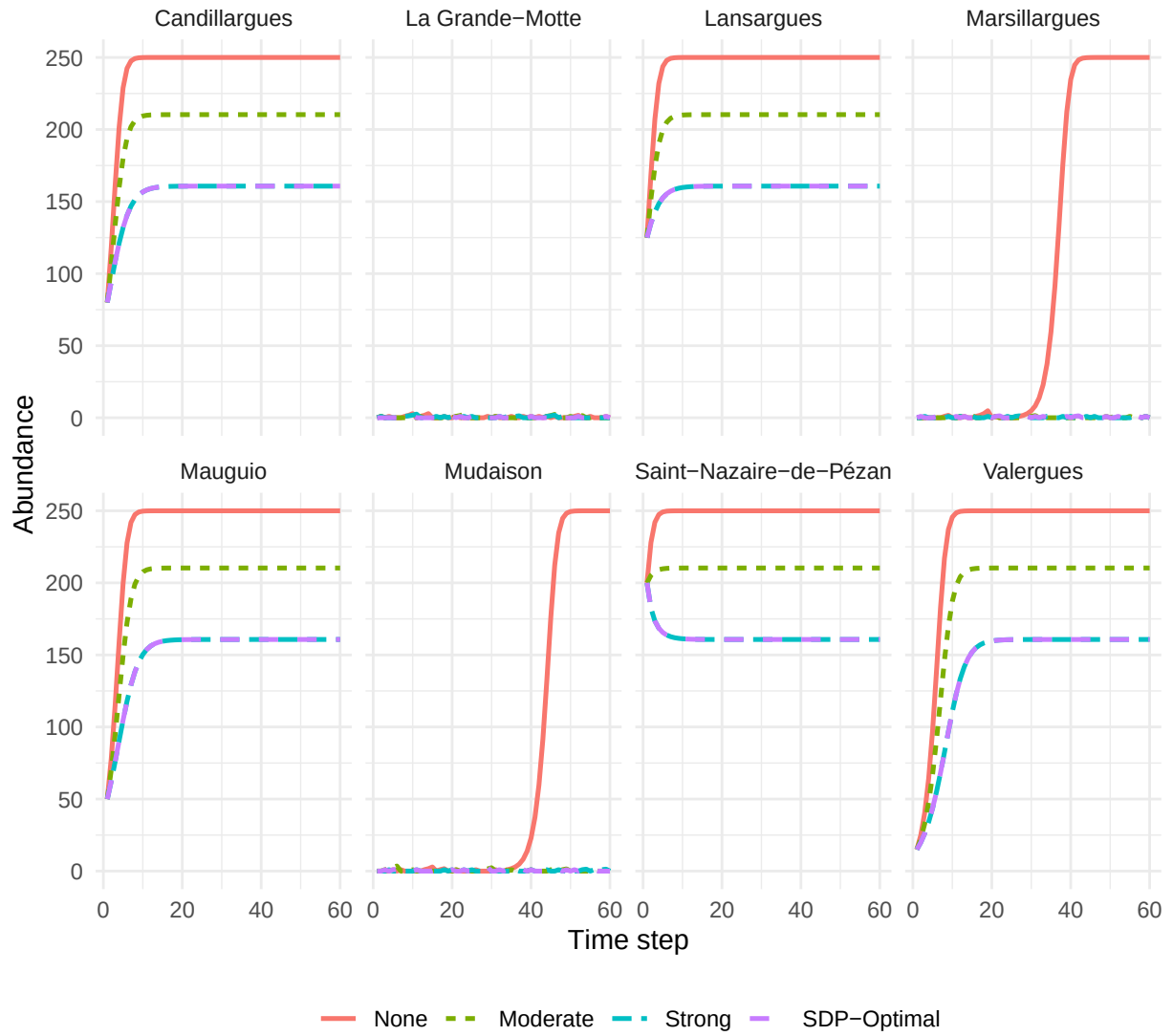


Figure 7: Site-level abundance dynamics under constant strategies and the SDP-optimal policy. Each panel corresponds to a site; trajectories highlight how spatial heterogeneity and network-driven recolonization can create uneven benefits of control across locations.

the system. However, POMDPs are more computationally and mathematically challenging to solve due to the continuous and high-dimensional state space. A detailed primer on developing POMDPs for ecological problems already exists (Chades et al 2021), so here we will keep our description simple.

POMDPs differ from MDPs by including an observation state (O), observation function (Z), and initial belief state (b_0). The observation state includes the set of observations perceived by the manager, while the observation function represents the conditional probability of a manager observing o' given that action a lead to state s' . In cases where state uncertainty cannot be reduced, the observation function is independent from the actions taken, and thus dependent only on state s' . The initial belief represents the believed state of the system at a given time.

Because the states are themselves unobservable, the state of the system must be tracked with belief states. At any point in time, the actual (unobservable) state of the system influences the immediate returns, state transitions, and observations, but do not influence actions. The belief state is updated based on observations, which subsequently inform the action(s) to be selected. Lastly, actions drive the state transitions, rewards, and may influence the observations.

[I think having a diagram to show the difference between MDP and POMDP, like Figure 1 in Chades et al 2021 or Fig 2-3 in Williams & Brown 2022, could be helpful] [Also from Chades et al 2021: “In the language of statistics, POMDPs are models that describe optimal control of hidden Markov models (HMMs). In this framing, the partially observed states of the model are latent variables that are inferred from the (fully observable) observation states.”]

Chades et al (2021) identified three types of management problems that POMDPs can be used to solve: - Exploiting current knowledge or reducing uncertainty: Classic dilemma where managers have to choose between investing in reducing state uncertainty or exploiting current knowledge. - Managing under imperfect detection: Assumes that managers have no means of reducing uncertainty, but managers can invest in management actions for which effectiveness

328 varies across the state space. - Adaptive management: Where the dynamics of the system
329 are not observable. The authors included an example for the first type of problem from a
330 conservation perspective. Here, we'll focus on an example for managing under imperfect
331 detection, continuing with our coyapu case study.

332 Going back to a simple example in invasive species management, imagine we have Occu-
333 pancy model with uncertain detectability

334 The invasive species can either be established or absent, with possible observations defined as
335 present or absent denoting if the species was observed in a given time step.

336 **Reducing model uncertainty w/ adaptive management (Abby)**

337 Decision theoretic methods can be used to make control decisions not only in the face of
338 uncertainty, but also to reduce uncertainty over time. Adaptive management, or “learning by
339 doing”, involves making decisions to achieve a management objective while simultaneously
340 learning about system dynamics to improve future management success ([Holling 1978](#), [Walters](#)
341 [1986](#), [Conroy and Peterson 2013](#)). Adaptive management approaches differ based on whether
342 they learn only from the past (passive adaptive management) or anticipate future learning
343 (active adaptive management) ([Chadès et al. 2017](#)).

344 Here we apply adaptive management strategies to inform control decisions for a single coyapu
345 population and to learn about and reduce uncertainty in the per-capita growth rate. We will
346 use the following discrete-time population model, where r is the intrinsic rate of growth; K
347 is the carrying capacity; $q(u)$ is the harvest rate as a function of action u ; R is the per-capita
348 growth rate; m describes whether the per-capita growth rate is linear, concave, or convex; and ϵ
349 is log-normally distributed noise:

$$N_{t+1} = N_t(1 + R(N_t)) - q(u) \times N + \epsilon$$

$$R(N_t) = r \times (1 - (N_t/K)^m)$$

350 If $m > 1$, most density—dependent change occurs at high population levels (close to the
 351 carrying capacity), which is theoretically common for species with life history strategies typical
 352 of large mammals. If $m < 1$, most density-dependent change occurs at low population levels,
 353 which is theoretically common for species with life history strategies typical of insects and some
 354 fishes (Fowler 1981).

355 We will consider a model set with five possible values of m . This model bounds, but does not
 356 contain, the true model we will use to simulate dynamics in the passive adaptive management
 357 approach (Figure 1A). These different values of m result in different optimal policies when
 358 system dynamics are known perfectly (Figure 1B).

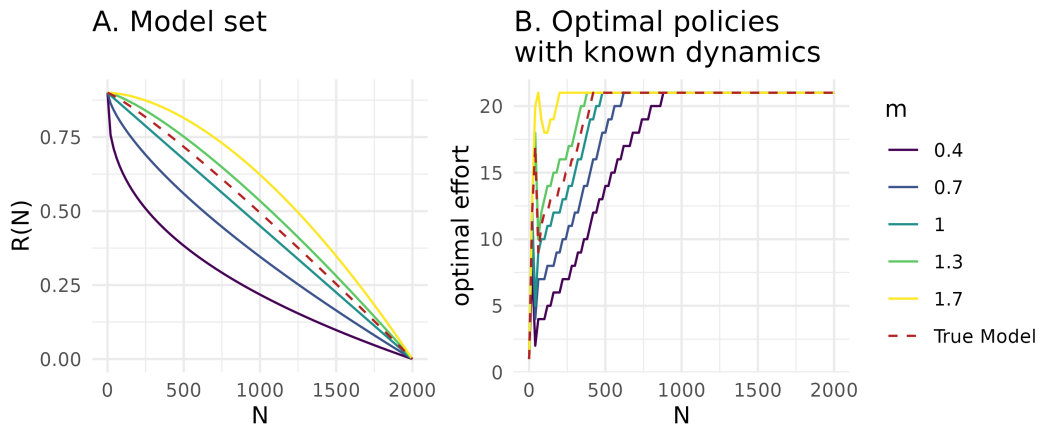


Figure 8: *A.* Model set representing structural uncertainty in the per-capita growth rate. *B.* Optimal state-dependent removal actions assuming perfectly known population dynamics, solved with stochastic dynamic programming. In both panels, colors correspond to different values of m in the model set, and the red dashed line corresponds to the true model dynamics used for simulation in the passive adaptive management approach.

Value of Information

For a decision maker, uncertainty may not be important to resolve; learning may not be inherently valuable if it does not result in improved management outcomes. A Value of Information (VoI) analysis *a priori* quantifies the importance of reducing uncertainty, or the expected improvement on management objectives if uncertainty were resolved (Raiffa and Schlaifer 2000). If the VOI is high, an adaptive management approach is typically justified.

The expected value of perfect information (EVPI) is measured as the difference between 1) PI , or the expected value once uncertainty has been resolved because the optimal action is chosen after knowing which model best describes the system, and 2) NL , or the expected value in the face of uncertainty because the optimal action is chosen with no learning about the system (Runge et al. 2011). The difference between these terms, $EVPI = PI - NL$, is the expected improvement in management performance due to acquiring perfect information.

To calculate EVPI for a sequential decision problem, we first use SDP to calculate the optimal state-dependent policy with assumed system dynamics, $\pi^*(s)_k$, for each model k in the model set K , given each transition matrix, P_k . We also calculate the bet-hedging strategy, or the optimal policy given uncertain dynamics, $\pi^*(s)_{NL}$. This bet-hedging strategy is calculated with the transition matrix P_{NL} , representing the average of the transitions of each model, P_k , weighted by each model belief, b : $P_{NL}(s_{t+1}|s_t, a_t) = \sum_{k \in K} b(k)P_k(s_{t+1}|s_t, a_t)$.

Using the policies calculated with known dynamics, $\pi^*(s)_k$, and unknown dynamics, $\pi^*(s)_{NL}$, we find the EVPI by first simulating i replicated population trajectories assuming each model k , applying both policies with known and unknown dynamics, and calculating the value at time t over time horizon T . The value associated with applying the policy with known dynamics is $V_{k,t,i}(s_t, a_t|a_t = \pi^*(s)_k)$, and the value associated with applying unknown dynamics is $V_{k,t,i}(s_t, a_t|a_t = \pi^*(s)_{NL})$. The EVPI is therefore $E_{s,k,t,i}[V(s_t, a_t|a_t = \pi^*(s)_k)] - E_{s,k,t,i}[V(s_t, a_t|a_t = \pi^*(s)_{NL})]$.

384 In the context of coypru regulation, the EVPI represents the expected improvement on the
 385 management objective if uncertainty about the per-capita growth rate was reduced. We calculate
 386 the EVPI for a model set, K , where $m \in \{0.4, 0.7, 1, 1.3, 1.7\}$ for $I = 200$ iterations over a $T = 40$
 387 time horizon. We assume equal belief in each model. The expected value once uncertainty has
 388 been resolved, PI , is -1730.4, the expected value with no learning, NL , is -1787.8, and the EVPI is
 389 $PI - NL = 57.4$. It is therefore valuable to learn about the population dynamics and per-capita
 390 growth rate, as this learning is expected to improve performance on the management objective.
 391 Value of information analyses are typically applied for static, one-time conservation decisions,
 392 rather than sequential problems, often in the context of understanding when to act and when to
 393 delay action until after further research (Bolam et al. 2019). To better understand the value of
 394 information in the context of adaptive management, future work should investigate quantifying
 395 VOI by iteratively resolving uncertainty, requiring evaluating the gain of implementing an
 396 adaptive policy rather than a single action (Chadès et al. 2017).

397 **Passive adaptive management**

398 Passive adaptive management involves learning from the outcomes of previous management
 399 actions to update knowledge about system dynamics and improve management performance
 400 over time. Decisions at each time step are optimized assuming that current knowledge of the
 401 system will not change into the future. Learning therefore occurs by monitoring the effect of an
 402 implemented action, rather than during the optimization procedure (Chadès et al. 2017).

403 Solving a passive adaptive management problem includes an implementation step and an
 404 updating step at every time point t , based on the current belief in model k , $b_t(k)$. The MDP must
 405 be solved at every time step because the transition matrices change as the belief changes. During
 406 the implementation step, the optimal action at time t is selected using a weighted averaging
 407 approach, where the transition probabilities are averaged across all models with the weights

408 given by the current belief, $b_t(k)$ (Williams et al. 2011). This can be solved using stochastic
 409 dynamic programming:

$$V^*(s_t|b_t) = \max_{a \in A} \left\{ \sum_{k \in K} b_t(k) r(s_t, a_t) + \gamma \sum_{s_{t+1}} \left\{ \sum_{k \in K} b_t(k) P_k(s_{t+1}|s_t, a_t) \times V^*(s_{t+1}|b_t) \right\} \right\}$$

410 During the updating step, the resulting state, s_{t+1} , after taking action a_t is observed, and belief,
 411 b_{t+1} , is updated according to Bayes' rule:

$$b_{t+1}(k|s_t, a_t, s_{t+1}, b_t) = \frac{P_k(s_{t+1}|s_t, a_t) b_t(k)}{\sum_{k \in K} P_k(s_{t+1}|s_t, a_t) b_t(k)}$$

412 For coyapu control decisions, we solve the passive adaptive management problem by updating
 413 the belief about the coyapu population dynamics to improve management performance over
 414 time. We begin with an equal belief, $b_{t=1}(k) = 1/5$, across the five models in the model set
 415 (Figure 1A). We iterate between 1) calculating the optimal action given the current belief state, 2)
 416 implementing the optimal action and simulating the next state, s_{t+1} , given the true population
 417 dynamics, and 3) updating the belief state based observing the next state given action a_t . Here
 418 we assume the state is perfectly observed. We simulate the application of passive adaptive
 419 management over 30 time steps. The belief in each model changes over time (Figure 2A),
 420 resulting in the highest belief for the two models closest to the true population dynamics $m = 1$
 421 and $m = 1.3$ (Figure 1A). Through updating the belief state over time and implementing the
 422 optimal action given the current belief state, the management performance improves over time
 423 (Figure 2B).

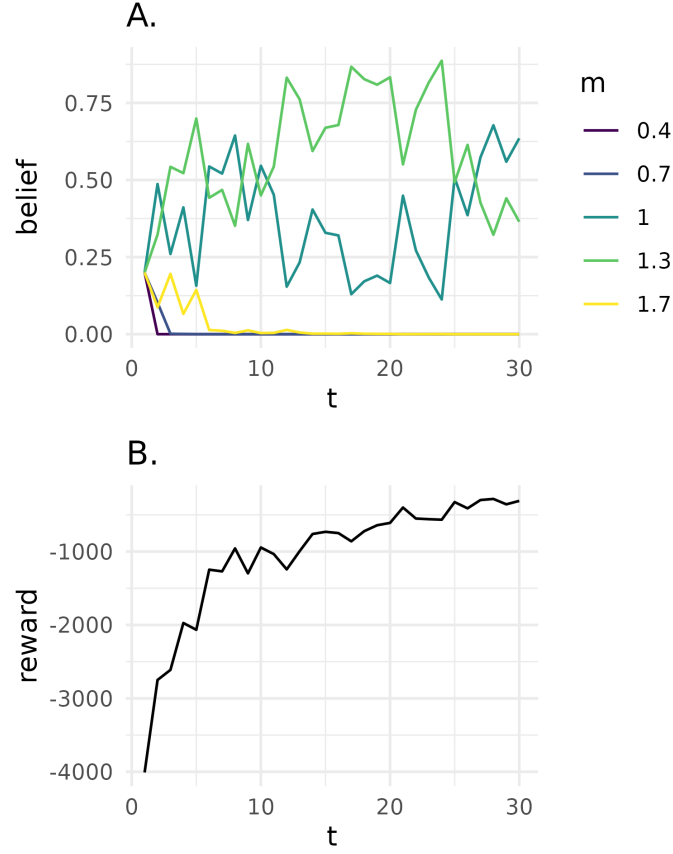


Figure 9: *A.* Belief state for each model k at time t . Colors correspond to different values of m , representing structural uncertainty in the per-capita growth rate. *B.* Reward, $r(s_t, a_t)$, at time t in passive adaptive management problem.

Active adaptive management

In contrast to passive adaptive management, active adaptive management does not rely only on past experience and instead explicitly accounts for future learning opportunities to maximize the chance of achieving the objective. Instead of an optimization at every time step as in passive adaptive management, the optimization occurs once and requires that the trajectory of belief b_t to b_{t+1} and state transitions be calculated for all time steps (Chadès et al. 2017). The optimal value function V^* that characterizes the performance of a policy is a function of both the state of the system s_t and belief over the models b_t :

$$V^*(s_t, b_t) = \max_{a \in A} \left\{ \sum_{k \in K} b_t(k) r(s_t, a_t) + \gamma \sum_{s_{t+1}} \left\{ \sum_{k \in K} b_t(k) P_k(s_{t+1} | s_t, a_t) \times V^*(s_{t+1}, b_t) \right\} \right\}$$

Active adaptive management with model uncertainty can be solved with many methods, including those developed for dealing with partial observability (Williams et al. 2011, Chadès et al. 2017). Here, a POMDP is characterized by a tuple $\langle X, A, O, P, Z, r, \gamma \rangle$, where $X = S \times K$ represents the factored state space with S denoting the possible conditions of the system and K denoting the model set. The unknown model is an unobserved state variable that must be inferred through observation. We can solve the POMDP with the Successive Approximations of the Reachable Space under Optimal Policies (SARSOP) algorithm (Kurniawati et al. 2008). With this algorithm, the value function V^* associated with the optimal policy π^* can be approximated by a convex, piecewise-linear function $V(b) = \max_{\alpha \in \Gamma} (\alpha \cdot b)$, where Γ is a finite set of α -vectors and b is the discrete vector representation of a belief (Kurniawati et al. 2008).

For coypru control decisions, we solve the active adaptive management problem by augmenting the state space with the belief in two models where $m \in \{0.7, 1.3\}$. We choose a smaller model set than in the passive adaptive management problem due to the computational cost associated with exploring the belief and state spaces over the entire time period (i.e., “curse of dimensionality”). We assume a uniform belief across models and assume perfect observation of the state (i.e., $o_t = s_t$).

We also use the coypru example to demonstrate the “dual control problem” (Walters and Hilborn 1978). A management action altering a poorly understood system can produce two types of benefits: 1) short-term payoffs and 2) learning about the system so as to improve payoffs in the long-term. There is a trade-off between these two benefits, as large disturbances allow for rapid learning yet may incur high costs; the challenge is to optimally balance these benefits. Here we explore the dual control problem by selecting two different discount factors, $\gamma \in \{0.95, 0.75\}$.

454 The higher of which represents a decision maker who values long-term rewards more highly,
 455 and the lower of which represents a decision maker who values short-term rewards more highly.
 456 The active adaptive management solution describes the optimal action (a_t , removal effort),
 457 given the state (s_t , coypu abundance), initial belief ($b_{t=1}$, uniform across models), and discount
 458 factor (Figure 3). The solution differs depending on how future rewards are discounted in the
 459 optimization. At low coypu abundance, the optimal removal effort increases as the discount
 460 factor increases. With a higher discount factor, future system states are valued more highly; even
 461 though a higher removal effort is costly today, this higher removal effort will accelerate learning
 462 to yield higher rewards in the long term (Figure 3A). In contrast, with a lower discount factor,
 463 active learning becomes less useful, as the price paid for not following the passive-adaptive
 464 (optimal right now) solution exceeds the information gained from learning (Figure 3B).

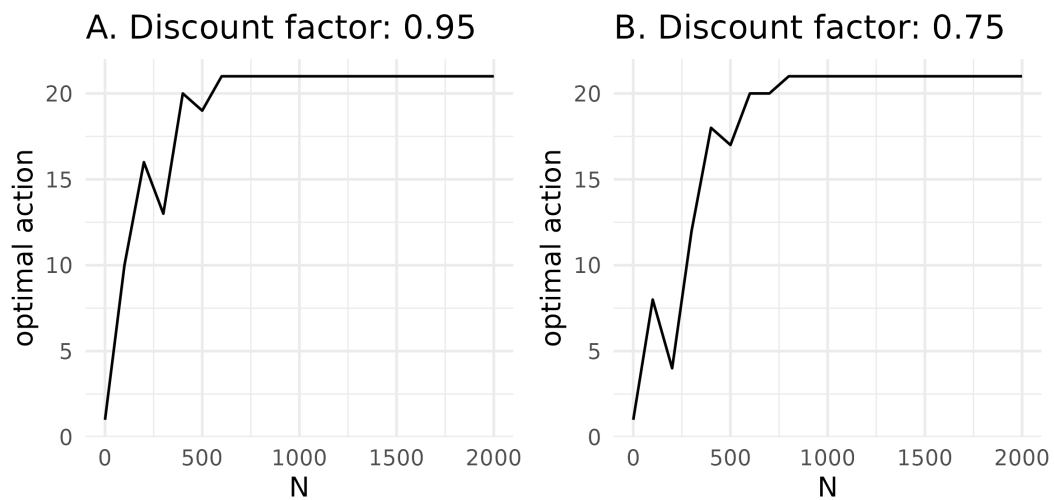


Figure 10: A. State-dependent active adaptive management solution, given an initial uniform belief across models and a higher discount factor representing a decision maker who values long-term payoffs more highly. B. State-dependent active adaptive management solution, given an initial uniform belief across models and a lower discount factor representing a decision maker who values short-term payoffs more highly.

465 On a single species ; check out Conroy's book, there is a nice figure illustrating AM.

466 <https://onlinelibrary.wiley.com/doi/10.1002/ece3.71176>

467 **Discussion (all)**

468 **Main messages**

469 How is SDM/SDP relevant.

470 **Limitations**

471 Curse of dimensionality.

472 **Perspectives**

473 **Acknowledgments**

474 **Data availability statement**

475 Data and code are available at <https://github.com/oliviergimenez/SDPaper>.

476 **References**

477 Bolam, F. C., M. J. Grainger, K. L. Mengersen, G. B. Stewart, W. J. Sutherland, M. C. Runge,
478 and P. J. McGowan. 2019. Using the value of information to improve conservation decision
479 making. *Biological Reviews* 94:629–647.

480 Chadès, I., S. Nicol, T. M. Rout, M. Péron, Y. Dujardin, J.-B. Pichancourt, A. Hastings, and C. E.
481 Hauser. 2017. Optimization methods to solve adaptive management problems. *Theoretical*
482 *Ecology* 10:1–20.

483 Conroy, M. J., and J. T. Peterson. 2013. Decision making in natural resource management: A
 484 structured, adaptive approach. John Wiley & Sons.

485 Fowler, C. W. 1981. Density dependence as related to life history strategy. *Ecology* 62:602–610.

486 Haubrock, P. J., T. Everts, N. A. S. Abreo, J. Bojko, V. Deklerck, J. W. E. Dickey, A. C. S. Franco,
 487 E. García-Berthou, S. Katsanevakis, N. I. Kirichenko, S. Mammola, M. A. Nuñez, B. Parker,
 488 R. Scalera, I. Soto, D. Strubbe, A. S. Tarkan, L. Vilizzi, T. Adriaens, P. Balzani, D. Blońska,
 489 E. Briski, R. Brys, A. L. Burgess, J. E. Byers, C. Cano-Barbacil, G. Castaldelli, J. T. A. Dick,
 490 V. Dominguez Almela, R. D. Dimarco, M. Florencio, A. Kouba, M. Kourantidou, I. Kurtul,
 491 I. Martín-Forés, O. Morissette, J. D. Olden, B. E. Soares, J. Truszkowski, H. Verreycken, M.
 492 Kenis, R. Sousa, and J. R. Britton. 2025. The impacts of biological invasions. *Biological*
 493 *Reviews* n/a.

494 Holling, C. S. 1978. Adaptive environmental assessment and management. John Wiley & Sons.

495 Kurniawati, H., D. Hsu, W. S. Lee, and others. 2008. Sarsop: Efficient point-based pomdp
 496 planning by approximating optimally reachable belief spaces. *Robotics: Science and systems*.
 497 Zurich, Switzerland.

498 Pyšek, P., P. E. Hulme, D. Simberloff, S. Bacher, T. M. Blackburn, J. T. Carlton, W. Dawson, F.
 499 Essl, L. C. Foxcroft, P. Genovesi, J. M. Jeschke, I. Kühn, A. M. Liebhold, N. E. Mandrak, L.
 500 A. Meyerson, A. Pauchard, J. Pergl, H. E. Roy, H. Seebens, M. van Kleunen, M. Vilà, M.
 501 J. Wingfield, and D. M. Richardson. 2020. Scientists’ warning on invasive alien species.
 502 *Biological Reviews* 95:1511–1534.

503 Raiffa, H., and R. Schlaifer. 2000. Applied statistical decision theory. John Wiley & Sons.

504 Roy, H. E., A. Pauchard, P. J. Stoett, T. Renard Truong, L. A. Meyerson, S. Bacher, B. S. Galil, P. E.
 505 Hulme, T. Ikeda, S. Kavileveettil, M. A. McGeoch, M. A. Nuñez, A. Ordonez, S. J. Rahlao, E.
 506 Schwindt, H. Seebens, A. W. Sheppard, V. Vandvik, A. Aleksanyan, M. Ansong, T. August,
 507 R. Blanchard, E. Brugnoli, J. K. Bukombe, B. Bwalya, C. Byun, M. Camacho-Cervantes, P.
 508 Cassey, M. L. Castillo, F. Courchamp, K. Dehnen-Schmutz, R. D. Zenni, C. Egawa, F. Essl,

G. Fayvush, R. D. Fernandez, M. Fernandez, L. C. Foxcroft, P. Genovesi, Q. J. Groom, A. I. González, A. Helm, I. Herrera, A. J. Hiremath, P. L. Howard, C. Hui, M. Ikegami, E. Keskin, A. Koyama, S. Ksenofontov, B. Lenzner, T. Lipinskaya, J. L. Lockwood, D. C. Mangwa, A. F. Martinou, S. M. McDermott, C. L. Morales, J. Müllerová, N. A. Mungi, L. K. Munishi, H. Ojaveer, S. N. Pagad, N. P. K. T. S. Pallewatta, L. R. Peacock, E. Per, J. Pergl, C. Preda, P. Pyšek, R. K. Rai, A. Ricciardi, D. M. Richardson, S. Riley, B. J. Rono, E. Ryan-Colton, H. Saeedi, B. B. Shrestha, D. Simberloff, A. Tawake, E. Tricarico, S. Vanderhoeven, J. Vicente, M. Vilà, W. Wanzala, V. Werenkraut, O. L. F. Weyl, J. R. U. Wilson, R. O. Xavier, and S. R. Ziller. 2024. Curbing the major and growing threats from invasive alien species is urgent and achievable. *Nature Ecology & Evolution* 8:1216–1223.

Runge, M. C., S. J. Converse, and J. E. Lyons. 2011. Which uncertainty? Using expert elicitation and expected value of information to design an adaptive program. *Biological Conservation* 144:1214–1223.

Thompson, B. K., J. D. Olden, and S. J. Converse. 2021. Mechanistic invasive species management models and their application in conservation. *Conservation Science and Practice* 3:e533.

Walters, C. J. 1986. *Adaptive management of renewable resources*. Macmillan Publishers Ltd.

Walters, C. J., and R. Hilborn. 1978. Ecological optimization and adaptive management. *Annual review of Ecology and Systematics* 9:157–188.

Williams, B. K., M. J. Eaton, and D. R. Breininger. 2011. Adaptive resource management and the value of information. *Ecological Modelling* 222:3429–3436.

Williams, B. K., J. D. Nichols, and M. J. Conroy. 2002. *Analysis and management of animal populations*. Academic Press, San Diego, California, USA.